# Report: Example D2

Date 2026-11-03  Engine FAKE fake-2  App 0.1.0

> ⚠ **FAKE engine:** These results come from the Fake Engine. They are placeholders, not engineering results.

## 1 Verdict

> ✕ **INFEASIBLE**: A valid operating point exists, but the design violates 2 constraint(s).

## 2 Limiting constraints

- Bus voltage, lower limit at B1: 0.9 pu against the limit 1 pu.
- Inverter loading at INV1: 1.1 ratio against the limit 1 ratio.

**Limiting constraints**

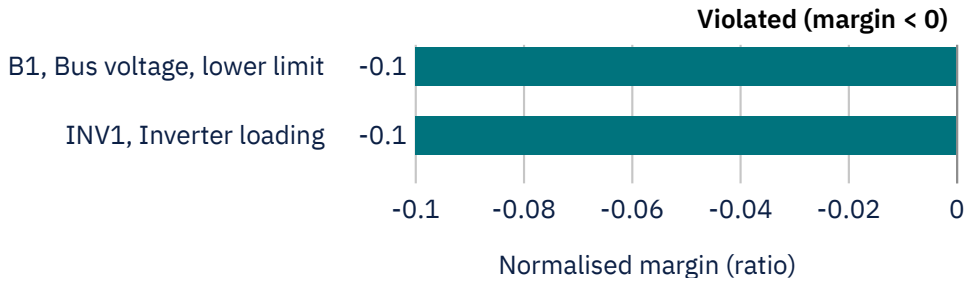
| Constraint | Element | Value | Limit | Unit | Margin |
|---|---|---|---|---|---|
| Bus voltage, lower limit | B1 | 0.9 | 1 | pu | ✕ **Violated by 0.1 pu** |
| Inverter loading | INV1 | 1.1 | 1 | ratio | ✕ **Violated by 0.1 ratio** |

## 3 Margins and checks not evaluated

**All evaluated margins**

| Constraint | Element | Value | Limit | Unit | Margin |
|---|---|---|---|---|---|
| Bus voltage, lower limit | B1 | 0.9 | 1 | pu | ✕ **Violated by 0.1 pu** |
| Inverter loading | INV1 | 1.1 | 1 | ratio | ✕ **Violated by 0.1 ratio** |

**Normalised margin per constraint**




12 check(s) without a limit are listed under Not evaluated, not drawn.

M2 · RunRecord 284e89fb4a7d4dafa52431315184845c · result hash fddd9d67397b677b172b87de8e8a1acc8cc177b76a8bc88d35a55dd6cfc12c33 · engine FAKE · **FAKE engine: placeholder, not an engine result**

## Not evaluated

- Bus voltage, lower limit, B1: no limit given; not counted as satisfied.
- Bus voltage, upper limit, B1: no limit given; not counted as satisfied.
- Bus voltage, lower limit, B2: no limit given; not counted as satisfied.
- Bus voltage, upper limit, B2: no limit given; not counted as satisfied.
- Bus voltage, lower limit, B3: no limit given; not counted as satisfied.
- Bus voltage, upper limit, B3: no limit given; not counted as satisfied.
- Frequency, lower limit, SYSTEM: no limit given; not counted as satisfied.
- Frequency, upper limit, SYSTEM: no limit given; not counted as satisfied.
- Inverter loading, INV1: no limit given; not counted as satisfied.
- Inverter loading, INV2: no limit given; not counted as satisfied.
- Line current, LN1: no limit given; not counted as satisfied.
- Line current, LN2: no limit given; not counted as satisfied.

## Loadability

| | |
|---|---|
| λ_c (constraint loadability, ratio) | not available |
| λ_s (ratio)<br>solution existence, not a stability limit | not available |
| Resolution (ratio) | 0.005 |
| Grid step (ratio) | 0.05 |
| Stop reason | FAKE_ENGINE |

**Loadability along the load scale λ**

No loadability value: λ_c, λ_s and λ_ub are not available.

Load scale λ (ratio)

● λ_c (solid) ■ λ_s (dashed) ▲ λ_ub (dotted)

Stop reason: FAKE_ENGINE. Bands: the bisection bracket of 0.005 above λ_c and λ_s. λ_s is a solution-existence diagnostic, not a stability limit. Not available: λ_c, λ_s, λ_ub.

M2 · RunRecord 284e89fb4a7d4dafa52431315184845c · result hash `fddd9d67397b677b172b87de8e8a1acc8cc177b76a8bc88d35a55dd6cfc12c33` · engine FAKE · **FAKE engine: placeholder, not an engine result**

Sizing check · from the network **Sizing check (advisory, from the network)**

| Inverter | S (kVA) | Required rating (kVA) | Headroom (kVA) | Rating margin short |
|---|---|---|---|---|
| **INV1** | 0 | 0 | 70 | no |
| **INV2** | 0 | 0 | 45 | no |

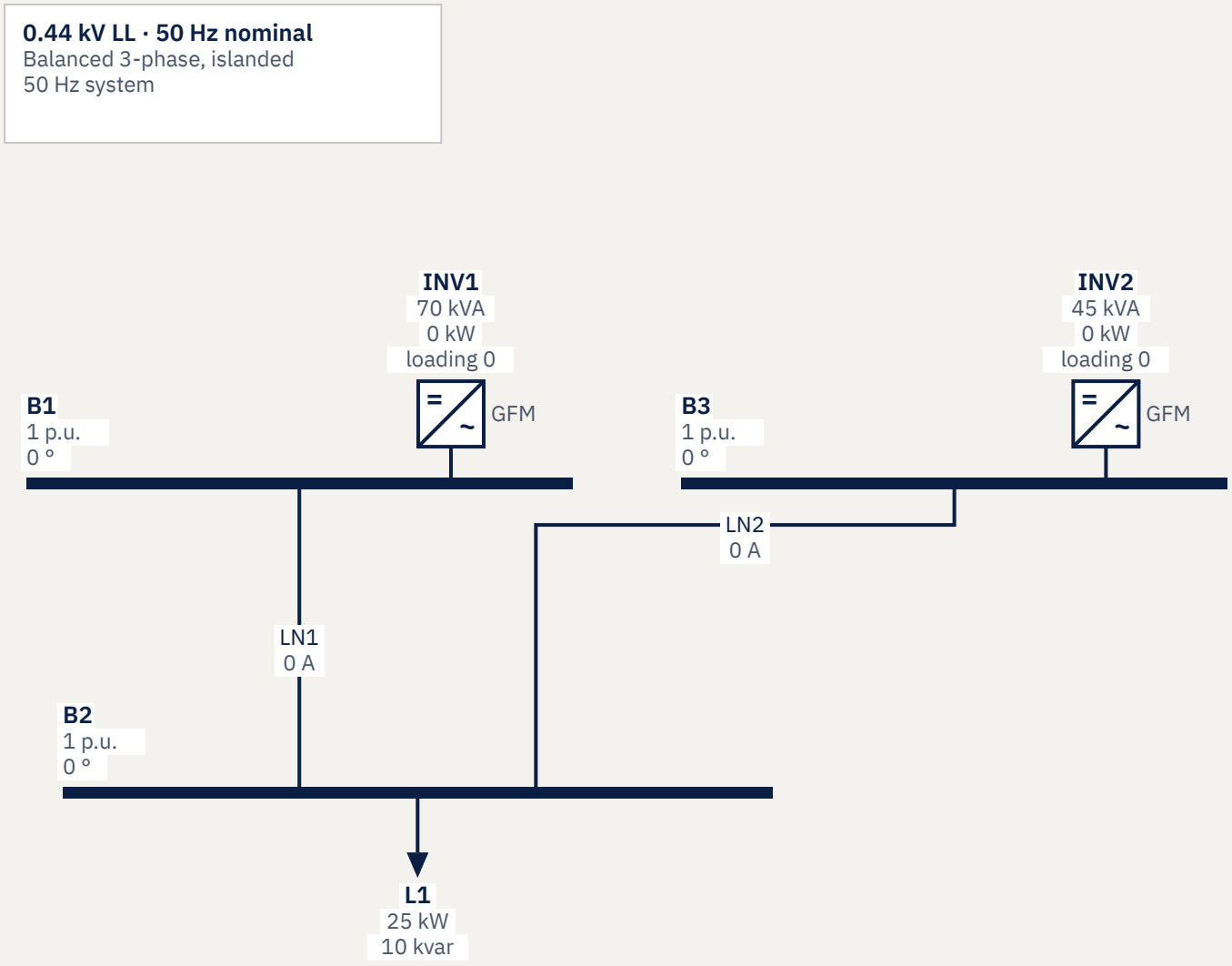
| Line | Current (A) | Required current (A) |
|---|---|---|
| **LN1** | 0 | 0 |
| **LN2** | 0 | 0 |

No storage check. Fake Engine placeholder; not a calculation result.

## 4 Design summary

| Quantity | Value | Unit |
|---|---|---|
| Nominal voltage, line-to-line | 0.44 | kV |
| Nominal frequency | 50 | Hz |
| Buses | 3 | count |
| Lines | 2 | count |
| Loads | 1 | count |
| Inverters | 2 | count |

## Single-line diagram of the analysed design

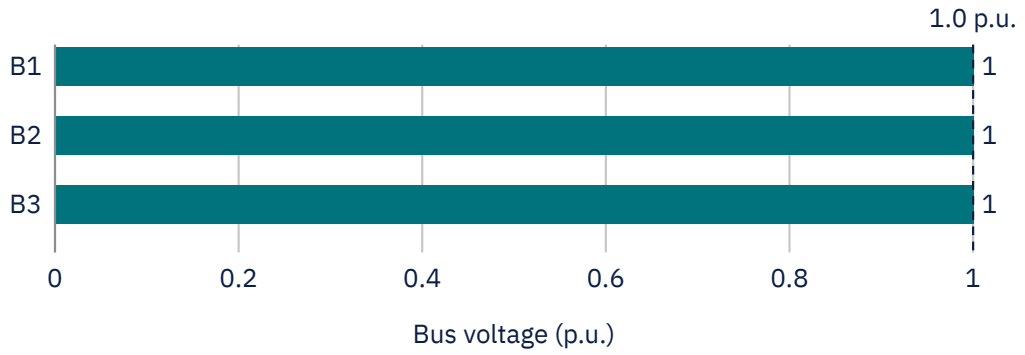



M1 · RunRecord 63eb41656d00448ba02ea73d9ac8105d · snapshot 149741dac4f0 · engine FAKE · **FAKE ENGINE — not an engine result**

Bus (busbar) · Grid-forming inverter · Load (constant PQ) · Line (series R–X)

# 5 Results and charts

**Bus voltage magnitude from the latest power flow (M1)**




M1 · RunRecord 63eb41656d00448ba02ea73d9ac8105d · result hash 4ff7a438f42d1cb61c2ad3ec6ea370d93a1e05c14494b078c77cfa27b48067bd · engine FAKE · **FAKE engine: placeholder, not an engine result**

**System**

| Quantity | Value | Unit |
|---|---|---|
| System frequency | 50 | Hz |
| Total load, active | 0 | kW |
| Total load, reactive | 0 | kvar |
| Total losses | 0 | kW |
| Iterations | 0 | count |
| Validity indicator ε_X | 0 | ratio |

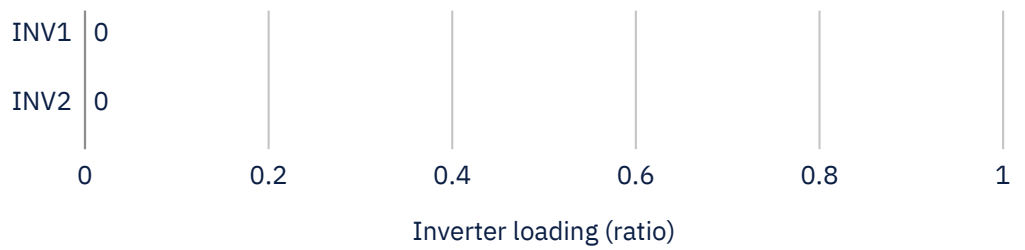
**Bus voltages**

| Bus | **Voltage** (kV) | **Voltage** (p.u.) | **Angle** (deg) |
|---|---|---|---|
| **B1** | 0.44 | 1 | 0 |
| **B2** | 0.44 | 1 | 0 |
| **B3** | 0.44 | 1 | 0 |

**Inverter outputs**

| Inverter | **P** (kW) | **Q** (kvar) | **S** (kVA) | **Loading** (ratio) |
|---|---|---|---|---|
| **INV1** | 0 | 0 | 0 | 0 |
| **INV2** | 0 | 0 | 0 | 0 |

### Inverter loading

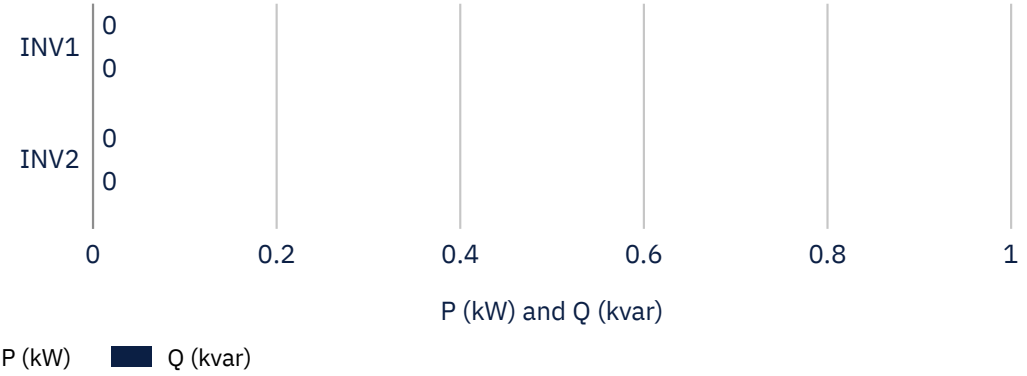



No inverter loading limit set: not evaluated.

M1 · RunRecord 63eb41656d00448ba02ea73d9ac8105d · result hash 4ff7a438f42d1cb61c2ad3ec6ea370d93a1e05c14494b078c77cfa27b48067bd · engine FAKE · **FAKE engine: placeholder, not an engine result**

Advice **Power sharing against rating-proportional sharing (advisory, not a constraint)**

### Inverter output, P and Q

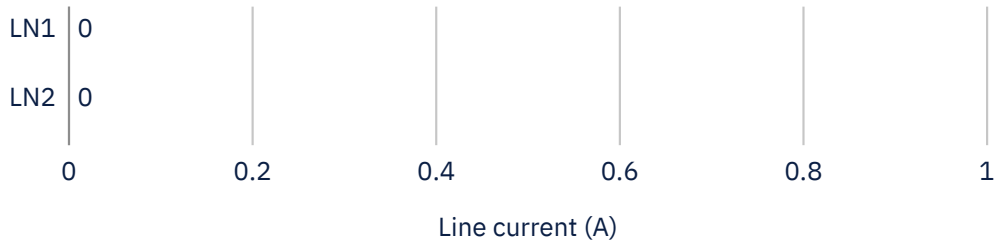



Deviations from rating-proportional sharing are in the table below.

M1 · RunRecord 63eb41656d00448ba02ea73d9ac8105d · result hash 4ff7a438f42d1cb61c2ad3ec6ea370d93a1e05c14494b078c77cfa27b48067bd · engine FAKE · **FAKE engine: placeholder, not an engine result**

| Inverter | **ΔP** (kW) | **ΔQ** (kvar) | **P error** (pu of rating) | **Q error** (pu of rating) | **Q–V droop not rating-proportional** |
|---|---|---|---|---|---|
| **INV1** | 0 | 0 | 0 | 0 | no |
| **INV2** | 0 | 0 | 0 | 0 | no |

**Line loading**




No line current limit set: not evaluated.

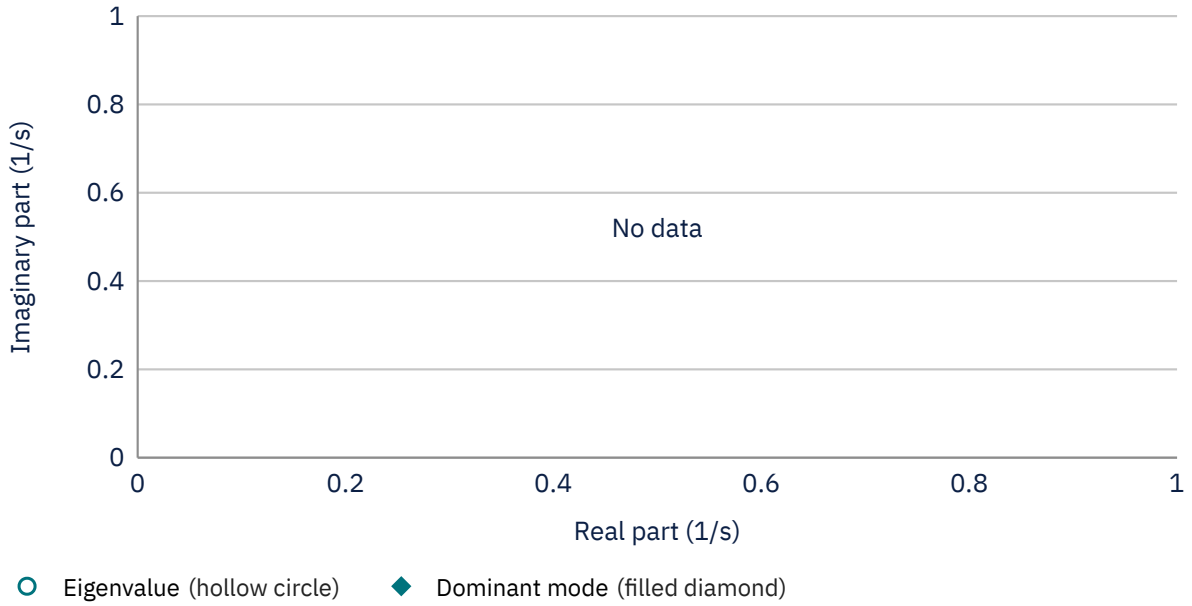
M1 · RunRecord 63eb41656d00448ba02ea73d9ac8105d · result hash 4ff7a438f42d1cb61c2ad3ec6ea370d93a1e05c14494b078c77cfa27b48067bd · engine FAKE · **FAKE engine: placeholder, not an engine result**

**Line flows**

| Line | P from (kW) | Q from (kvar) | P to (kW) | Q to (kvar) | Current (A) | Losses (kW) |
|---|---|---|---|---|---|---|
| **LN1** | 0 | 0 | 0 | 0 | 0 | 0 |
| **LN2** | 0 | 0 | 0 | 0 | 0 | 0 |

## Small-signal (M4a): class STABLE

**Eigenvalues in the complex plane**




Hover a point for its damping ratio and frequency; the table lists every mode. No damping-threshold line is drawn.

M4a · RunRecord 62a69f7a67cb4993bec48a0690eb2793 · result hash b39d4ca4fd850387ee962830a6bf4f60a79239ae3c7cf6cb43272e2d64a19fc7 · engine FAKE · **FAKE engine: placeholder, not an engine result**

Participation factors not available: not reported.

# 6 Assumptions

Sizing check: Fake Engine placeholder; not a calculation result.

**Validity indicators: threshold not yet calibrated**

| Indicator | Checks assumption | Value (ratio) | From |
|---|---|---|---|
| r_τ line-to-filter time ratio (Eq. 4.23) | A1 | not evaluated: FAKE_ENGINE | M4a run 62a69f7a67cb4993bec48a0690eb2793 (Fake Engine) |
| κ dominant-mode speed (Eq. 4.23) | A1 | not evaluated: FAKE_ENGINE | M4a run 62a69f7a67cb4993bec48a0690eb2793 (Fake Engine) |
| r_in inner-loop ratio (Eq. 4.23) | A2 | not evaluated: FAKE_ENGINE | M4a run 62a69f7a67cb4993bec48a0690eb2793 (Fake Engine) |
| ε_X validity indicator (Eq. 4.1) | A4 | not evaluated: FAKE_ENGINE | M4a run 62a69f7a67cb4993bec48a0690eb2793 (Fake Engine) |

**Model assumptions (manuscript Tabel 1.1) and how each is handled**

| Assumption | Statement | Handling | Indicator values | RunRecord result hash |
|---|---|---|---|---|
| A1 | Jaringan kuasi-statis (saluran dimodelkan aljabar) | Checked by indicator (r_τ, κ) | r_τ: not evaluated: FAKE_ENGINE; κ: not evaluated: FAKE_ENGINE | `b39d4ca4fd850387ee962830a6bf4f60a79239ae3c7cf6cb43272e2d64a19fc7` |
| A2 | Inverter sebagai sumber tegangan ideal inner loop dalam, filter LC, PLL, dan pembatas arus | By construction; checked by indicator (r_in) | r_in: not evaluated: FAKE_ENGINE | `b39d4ca4fd850387ee962830a6bf4f60a79239ae3c7cf6cb43272e2d64a19fc7` |
| A3 | Konstanta waktu filter daya ditetapkan pengguna | Set by the user; explored by sweep (M3b) | | |
| A4 | Reaktansi saluran tidak diperbarui terhadap frekuensi; tanpa shunt | Checked by indicator (ε_X) | ε_X: not evaluated: FAKE_ENGINE | `b39d4ca4fd850387ee962830a6bf4f60a79239ae3c7cf6cb43272e2d64a19fc7` |
| A5 | Kendali droop primer saja | By construction | | |
| A6 | Tiga fasa seimbang | By construction | | |

# 7 Traceability

| Analysis | RunRecord id | Snapshot hash | Result hash |
|---|---|---|---|
| M1 | 63eb41656d00448ba02ea73d9ac8105d | `149741dac4f0e99f80aa65b8353605b957f1e08ba2402395ad11f319e12b1288` | `4ff7a438f42d1cb61c2ad3ec6ea370d93a1e05c14494b078c77cfa27b48067bd` |
| M2 | 284e89fb4a7d4dafa52431315184845c | `149741dac4f0e99f80aa65b8353605b957f1e08ba2402395ad11f319e12b1288` | `fddd9d67397b677b172b87de8e8a1acc8cc177b76a8bc88d35a55dd6cfc12c33` |
| M4a | 62a69f7a67cb4993bec48a0690eb2793 | `149741dac4f0e99f80aa65b8353605b957f1e08ba2402395ad11f319e12b1288` | `b39d4ca4fd850387ee962830a6bf4f60a79239ae3c7cf6cb43272e2d64a19fc7` |